# AI by Manvendra

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## 🔖 AI Topics – 2-Liner Summary Sheet

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### **1.** 🤖 Intelligent Agents

An AI agent perceives its environment and takes actions to achieve goals.

Types include reflex, model-based, goal-based, utility-based, and learning agents.

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# 2. 🔍 Search Algorithms

Search helps agents find action sequences to reach a goal.
Uninformed (BFS, DFS, UCS) and Informed (A*, Greedy) strategies explore state spaces.

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# 3. 🔼 Local Search

Optimizes by moving to neighboring states without building a full tree.
Includes Hill Climbing, Simulated Annealing, and Genetic Algorithms.

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# 4. 🧩 Constraint Satisfaction Problems (CSPs)

CSPs involve assigning values to variables under constraints (e.g., Sudoku).
Solved via backtracking, constraint propagation, and heuristics like MRV/LCV.

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# 5. ✳️ First-Order Logic (FOL)

FOL uses predicates, quantifiers, and logical inference for complex knowledge.
It supports variables and relations unlike propositional logic.

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# 6. 🧠 Knowledge-Based Agents

These agents use a knowledge base and inference to decide actions.
They separate knowledge from logic rules and use entailment for reasoning.

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## 7. 🎮 Adversarial Search & Games

Used in competitive environments (e.g., chess) with opponents.
Minimax and Alpha-Beta pruning select optimal moves against intelligent adversaries.

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## 8. 📦 Automated Planning

Planning is the process of creating action sequences to reach goals.

It uses STRIPS-like action models with preconditions and effects.

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## 9. 🌀 Markov Decision Processes (MDPs)

MDPs model decision-making under uncertainty using states, actions, and rewards.

Value iteration and policy iteration compute optimal strategies.

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## 10. 🧪 Reinforcement Learning (RL)

RL lets agents learn by interacting with the environment and receiving rewards.

Q-learning and SARSA help find optimal policies without a model.

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## 11. 📊 Bayesian Networks

Probabilistic models that represent dependencies using a directed graph.

Use Bayes' theorem and conditional probability for inference.

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## 12. 🧠 Machine Learning

ML enables systems to learn from data and make predictions.

Includes supervised, unsupervised, and deep learning approaches.